Requirements Document

Active Learning Platform for Selecting Data for Human Labeling

DEPI - AWS Machine Learning Engineer Track | Graduation Project
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1. Purpose and Scope

This document defines the requirements of the Active Learning Platform: a cloud-based system on AWS that trains an image classifier, selects the most informative unlabeled images, collects human labels for them, and retrains in repeated rounds. It is the basis for the analysis and design phase. Priorities follow the MoSCoW scale (Must, Should, Could).

2. Stakeholders and Users

Stakeholder	Role in the system	Main need
Annotator (team member)	Labels images selected by the system	Simple, fast labeling page; clear task list
ML engineer	Configures and runs experiments	Swappable query strategies; reproducible runs; metrics
Cloud administrator	Manages AWS resources	Low cost, secure access, monitoring
Project supervisor / DEPI evaluators	Assess the project	Working demo, results, documentation on time

3. Requirements Sources

- DEPI project documentation guidelines and deadlines.
- The literature review (active learning strategies, challenges and baselines).
- AWS documentation of the candidate services, including the closure of SageMaker Ground Truth to new customers.
- Team discussions to agree on scope, roles and budget.

4. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall store the image dataset in Amazon S3, separated into labeled, unlabeled and test sets.	Must
FR-02	The system shall create an initial labeled seed set and an unlabeled pool from the dataset.	Must
FR-03	The system shall train an image classification model on the current labeled set.	Must
FR-04	The system shall evaluate the model on a held-out test set and record accuracy per iteration.	Must
FR-05	The system shall run inference on the unlabeled pool and produce class probabilities.	Must
FR-06	The system shall compute uncertainty scores using least confidence, margin and entropy, and support random selection as a baseline.	Must
FR-07	The system shall select the top N samples according to the chosen strategy, with N configurable.	Must

ID	Requirement	Priority
FR-08	The system shall create labeling tasks for the selected samples.	Must
FR-09	The system shall provide a labeling interface where an annotator views an image, selects a class and submits.	Must
FR-10	The system shall store submitted labels with annotator, timestamp and iteration number.	Must
FR-11	The system shall add new labels to the labeled set and trigger retraining.	Must
FR-12	The system shall run the loop for a configurable number of iterations with minimal manual steps.	Should
FR-13	The system shall provide a dashboard showing accuracy, labels used and cost per iteration.	Should
FR-14	The dashboard shall compare strategies against the random baseline on the same chart.	Should
FR-15	The system shall log the configuration, random seed and model version of every iteration.	Should
FR-16	The system could support additional strategies such as MC dropout or ensembles.	Could

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Inference latency under 1 second per image for the deployed model.
NFR-02	Availability	Endpoint and dashboard available at least 99% of the time during demo and testing.
NFR-03	Cost	Total AWS spend within the agreed budget, with budget alerts and endpoints stopped when idle.
NFR-04	Security	Private S3 buckets, IAM roles with least privilege, no credentials in source code, authenticated access to the labeling app.
NFR-05	Reproducibility	Fixed random seeds, versioned datasets and models, and logged configurations.
NFR-06	Modularity	Query strategies and models are interchangeable without changing the rest of the pipeline.
NFR-07	Usability	Labeling page works on desktop and mobile browsers; labeling one image takes at most two actions.
NFR-08	Scalability	Handles an unlabeled pool of at least 50,000 images.
NFR-09	Observability	Pipeline steps log to CloudWatch so failures can be traced.

6. Use Cases

Use case	Actor	Summary
Run an active learning iteration	ML engineer	Starts the loop; system trains, scores, selects and creates labeling tasks
Label images	Annotator	Opens the labeling page, labels the assigned images, submits
Monitor results	ML engineer / Supervisor	Views accuracy, labels used and cost per iteration on the dashboard
Compare strategies	ML engineer	Runs the same budget with different strategies and compares curves
Manage resources and cost	Cloud administrator	Sets budget alerts, starts and stops endpoints, reviews logs

7. Data Requirements (proposed starting values)

- Dataset: CIFAR-10 (60,000 images, 10 classes), with the standard 50,000 training and 10,000 test split.
- Initial labeled seed set: a small random subset (for example 1,000 images, balanced by class), to be tuned after the first baseline run.
- Query batch size per iteration: for example 500 images; number of iterations: at least 5.
- Label format: image identifier, class, annotator, timestamp, iteration.

8. Constraints and Assumptions

- Project deadlines set by DEPI: design on 6 Nov 2026, implementation on 30 Nov 2026, final delivery on 4 Dec 2026.
- Amazon SageMaker Ground Truth is closed to new customers (effective 30 July 2026), so labeling does not depend on it.
- The team has four members and uses a limited AWS budget or program credits.
- Annotators are team members, so labeling quality is assumed high; a public dataset provides reference labels for evaluation.
- Only image classification is addressed.

9. Acceptance Criteria

Criterion	Linked requirements / KPIs
The loop runs for at least 5 consecutive iterations without manual fixes	FR-03 to FR-12; pipeline reliability KPI
At least one strategy beats random sampling at the same label budget	FR-06, FR-07, FR-14; gain over random KPI
Final model reaches at least 90% of the fully supervised accuracy using 30% or fewer labels	FR-04; model accuracy and labeling efficiency KPIs
Dashboard shows accuracy, labels used and cost per iteration	FR-13, FR-14
Inference under 1 second and uptime of 99% during demo	NFR-01, NFR-02
All deliverables submitted on schedule	Schedule KPI